

Diverse Envolving Worlds

A Standard Simulation Worlds



HM3D



MP3D

B Photorealistic Simulation



Habitat-GS



Interior-GS

C Evolving Simulation Worlds



HSSD

D Real-World Evolution



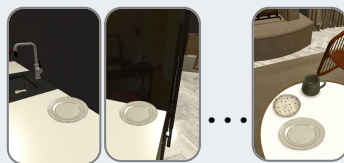
Indoor



Outdoor

Memory
Construction

1 Observed History



t_1

t_2

t_k

2 Hidden Evolution



world evolves

3 Current Belief



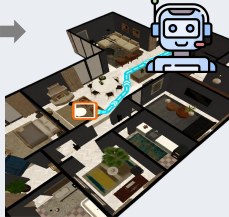
state₁

state₂

state_k

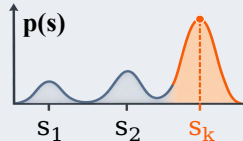
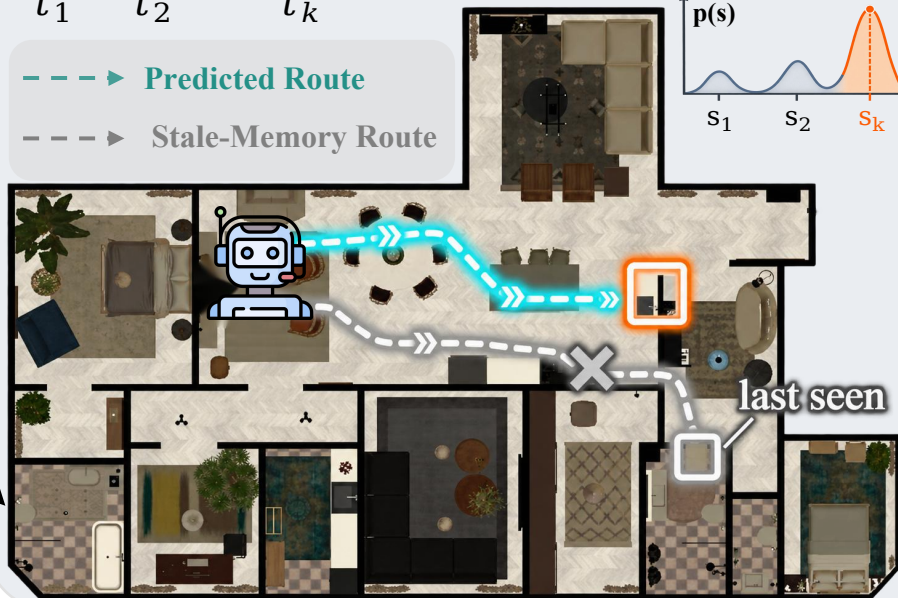
Memory
Update

4 Belief-guided Plan/Act



---> Predicted Route

---> Stale-Memory Route



P4D-Bench



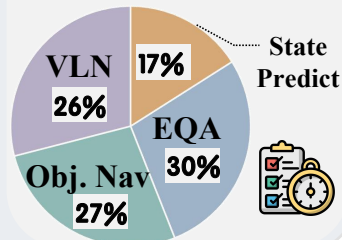
54 Scenes

803.68k Tasks

4860 Scene-Days

239.19k Dynamic

State Changes



Unified Embodied Tasks

A Embodied QA



Q: Where did I put the cup after I left the dining table ?

B Active Exploration



C Predictive Navigation



D Real World Interaction



Task: Find the lounge on this floor and enter it.